

DIGITAL MEDIA

FORENSIC INVESTIGATION

DIGITAL MEDIA FORENSIC REPORT

Forensic Media Investigation

CASE	CASE-20260902-111125-62A620
EVIDENCE	TEST-04-appended.png
FILE TYPE	PNG image data, 1280 x 959, 8-bit/color RGB, non-interlaced
FILE SIZE	1.19 MB
SHA-256	72a54d90479d56772c5429717a2580a24b6413ad62b576c2fc75fe15d83c3fb7

NOT VERIFIED

No verified hidden-data recovery established

SUSPICIOUS_IN
DICATORS_PRE
SENT
FINAL ASSESSMENT

LOW

RISK

MEDIUM

CONFIDENCE

The investigation did not establish verified hidden-data recovery from the supplied evidence.

01 File Identification

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02 Triage

Printable strings identified	15641
Raw string output	Preserved in structured analysis artifacts; not reproduced in this report.

03 Metadata Forensics

Assessment	NOTABLE METADATA PRESENT
GPS present	No
XMP present	Yes
Editing software detected	No
Comments present	No
Author present	No

04 Structure Analysis

Format	PNG
Valid signature	Yes
Trailing data	30 bytes
Embedded file signatures	None detected
Assessment	STRUCTURAL ANOMALY REQUIRES REVIEW

05 Visual & Bit-Plane Analysis

Channel	Mean	Std. deviation	LSB one-ratio
Red	4.5992	14.5671	0.2180
Green	24.8528	56.1803	0.2550
Blue	16.4408	43.6391	0.2343

RGB channel intensity distributions differ The green channel has the highest mean intensity (24.8528), while the red channel has the lowest mean intensity (4.5992).

LSB distributions differ across RGB channels The green channel has the highest LSB one-ratio (0.2550), while the red channel has the lowest LSB one-ratio (0.2180).

Bit-plane entropy varies across channels and positions Measured bit-plane entropy varies across RGB channels and bit positions. These measurements describe the observed pixel distribution and are not, by themselves, evidence of hidden data.

06 Statistical Analysis

Channel	LSB one-ratio	Entropy
Red	0.2180	2.7306
Green	0.2550	3.9636
Blue	0.2343	3.8504

The green channel has the highest LSB one-ratio (0.254993), while the red channel has the lowest LSB one-ratio (0.217983).

The green channel has the highest measured entropy (3.963590), while the red channel has the lowest measured entropy (2.730631).

LSB one-ratios vary across image regions. The regional spread describes spatial variation in the measured LSB distribution and does not independently establish steganography.

Interpretation: Statistical and LSB measurements are supporting observations and are not independently treated as proof of hidden data.

07 Automated Steganalysis

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CANDIDATE OBSERVATIONS

Automated steganalysis results are treated as candidate observations. Confirmation is established separately through the verification stage.

08 Verification & Extraction

Verification status	NOT VERIFIED
Technique	None
Verification score	None
Extraction successful	No
Recovered payload	Not available

NO VERIFIED RECOVERY

Extraction attempts did not establish verified hidden-data recovery.

Steghide: Extraction was unsuccessful under the tested conditions; this does not establish absence of hidden data.

09 Payload Analysis

Payload status	NOT IDENTIFIED
Classification	NOT_AVAILABLE
File type	NOT_AVAILABLE
Size	0 bytes

Payload SHA-256	NOT_AVAILABLE
Magic signature	Unknown / no known signature
Printable strings	0
YARA matches	0
Archive container	No

10 YARA Detection

NO YARA AVAILABLE	0 RULES LOADED	0 MATCHES
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NO KNOWN YARA INDICATORS DETECTED

11 Evidence Correlation, Decision & Risk

0 CORRELATION GROUPS	0 RISK SCORE	YES FURTHER ANALYSIS
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Evidence was normalized successfully, but no multi-item corroboration was established. Observations, indicators, and inconclusive results remain individually classified.

Decision

Final decision	SUSPICIOUS_INDICATORS_PRESENT
Further analysis required	Yes

Decision rationale

- One or more investigative indicators were observed, but no finding reached VERIFIED state.
- These indicators require contextual analysis and must not be treated as proof of hidden data or maliciousness.

Risk rationale

- No evidence met the threshold for a positive risk contribution.

12 Investigator Conclusion

- The investigation identified suspicious forensic indicators, but the supplied evidence did not establish verified hidden-data recovery.
- The evidentiary risk assessment was LOW with a score of 0.
- Further forensic analysis is recommended based on the existing evidence and decision-layer assessment.
- The findings describe the forensic evidence recovered during analysis and do not by themselves establish malicious intent.

Limitations

- Candidate evidence represents an investigative lead and should not be treated as verified hidden content.
- Plausible findings remain below the verification threshold and require additional validation.
- YARA matches are contextual indicators and do not automatically establish maliciousness.

- The reported risk score represents evidentiary suspicion and concealed-content significance; it is not a probability of maliciousness.
- The decision layer identified a need for further analysis of the available evidence or recovered artifacts.

Full machine-readable module results and raw tool artifacts are preserved separately. This PDF presents only selected evidence required for investigator interpretation and case-level reporting.